

# AVIN P S

## Data Scientist / ML Engineer

📍 Thrissur, Kerala, India

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### Summary

Final year Computer Science & Engineering student (Data Science) with hands-on experience designing and deploying end-to-end AI solutions. Proficient in Python, Machine Learning, and Generative AI, with live, real-world deployed projects and a proven ability to build, document, and communicate data-driven solutions.

### Skills

**Languages:** Python (Advanced), SQL, JavaScript/TypeScript (Basic)    **ML & AI:** Scikit-Learn, Gradient Boosting, SHAP, GenAI, LLM Integration

**Deep Learning & CV:** OpenCV, YOLO, CNN pipelines    **Data Science:** Pandas, NumPy, EDA, Statistics, Hypothesis Testing

**Visualization:** Matplotlib, Seaborn, Plotly Dash, Folium    **Deployment:** FastAPI, React, Streamlit, Vercel, Git

### Projects

**AI Personal Health Coach** | *Python, FastAPI, React, Scikit-Learn, SHAP, Render*    **Feb 2026 – Present**

- **Architected and deployed a production-grade AI health risk platform (Phase 1 live):** 7 Gradient Boosting Classifiers output calibrated 0–100 risk scores across Diabetes, Cardiovascular, Hypertension, Obesity, Respiratory, Mental Health, and Physical Health domains, validated on a custom 75,000-sample synthetic dataset.
- Integrated **per-sample SHAP** (TreeExplainer) with semantic suppression logic, dynamically filters irrelevant risk drivers per user, delivering plain-English, clinically actionable explanations with every prediction.
- Executing a **5-phase roadmap:** Phase 2 (K-Means lifestyle clustering, JWT auth, SQLAlchemy DB) → Phase 3 (AI meal & workout plans) → Phase 4 (adaptive feedback loop, plateau detection) → Phase 5 (personalized AI health chatbot on user's longitudinal data).

**KTU Result Analyser** | *React, TypeScript, PDF Parsing, Gemini API, Vercel*    **Dec 2025 – Feb 2026**

- **Deployed full-stack result analysis platform** actively used by IES College of Engineering's management and faculty and **adopted by multiple colleges across Thrissur district**, for institutional academic performance tracking and decision-making.
- Built a PDF parsing pipeline with Gemini LLM integration for automated subject-credit extraction and SGPA calculation; Department-Wise Report feature enables downloadable cohort-level analysis for faculty.

**SpaceX Falcon 9 Landing Prediction (IBM Capstone)** | *Python, SQL, Scikit-Learn, Plotly Dash*    **Jan 2026**

- **End-to-end ML pipeline**, multi-source data integration (SpaceX API + web scraping), SQL/Pandas EDA documented as technical reports, 4 classification models hyper-tuned to **83% accuracy (SVM & KNN)**.
- Designed interactive geospatial analytics (Folium) and a real-time Plotly Dash dashboard to communicate launch outcomes across global sites to non-technical stakeholders.

### Awards & Certifications

- **IBM Data Science Professional Certificate – Coursera (2026):** 12 hands-on courses: Python, SQL, EDA, Machine Learning, **Generative AI**, and Capstone project directly aligned with industry data science workflows.

### Leadership & Extracurricular

- **Chief Editor, DataXplorer** (Dept. Data Science Newsletter, 2024–Present) — leads editorial planning, content curation, and publication, communicating research trends and industry developments to the academic community.
- **Coordinator, College Technical & Arts Fests** — IES College of Engineering; managed cross-functional teams and end-to-end execution of college-wide events under high-pressure, time-bound conditions.

### Education

**B.Tech Computer Science & Engineering (Data Science)**

*APJ Abdul Kalam Technological University, IES College of Engineering, Thrissur, Kerala*

**Graduation: 2027**

*CGPA: 8.0 / 10*